

Multiple Choice (10 marks)

1. A _____ is a method in which a computer security mechanism is bypassed untraceable for accessing the computer or its information.
 - (a) front-door
 - (b) backdoor
 - (c) clickjacking
 - (d) key-logging
2. The stack is memory for storing
 - (a) Local variables
 - (b) Program code
 - (c) Dynamically linked libraries
 - (d) Global variables
3. What is penetration testing?
 - (a) A procedure for testing libraries or other program components for vulnerabilities
 - (b) Whole-system testing for security flaws and bugs
 - (c) A security-minded form of unit testing that applies early in the development process
 - (d) All of the above
4. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
 - (a) No, I cannot compute the key.
 - (b) Yes, the key is $k = m \text{ xor } c$.
 - (c) I can only compute half the bits of the key.
 - (d) Yes, the key is $k = m \text{ xor } m$.
5. Which among them has the strongest wireless security?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3

True / False (10 marks)

Answer True or False

6. True or False: Troya is classified as a remote Trojan, known for its ability to control infected devices from a distance.
7. True or False: A Base Signal Station is primarily used for data transmission in mobile networks, similar to a Base Transceiver Station.
8. True or False: The Tor browser is essential for accessing the Tor network and utilizing its anonymous services.
9. True or False: The Heartbleed vulnerability enables the execution of a format string attack, compromising system integrity.
10. True or False: The Morris Worm, released in 1988, is considered the first buffer overflow attack in computer security history.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss why traditional anti-virus scanners are ineffective against vulnerabilities like Heartbleed, focusing on the nature of the exploit and the limitations of signature-based detection methods.
12. Discuss the role of address sanitizer (ASAN) in identifying memory errors during fuzz testing of a program. How does ASAN enhance error detection and analysis?

End of Paper